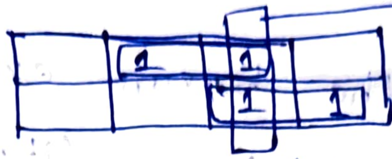
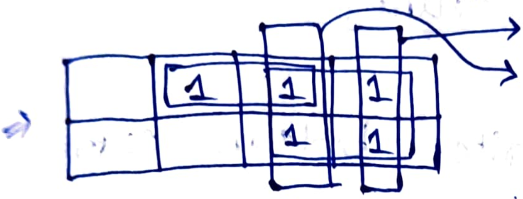
N) Make the best Possible Pairs. Avoid redundant Pairs. No. of Pairs should be as minimum as Possible.

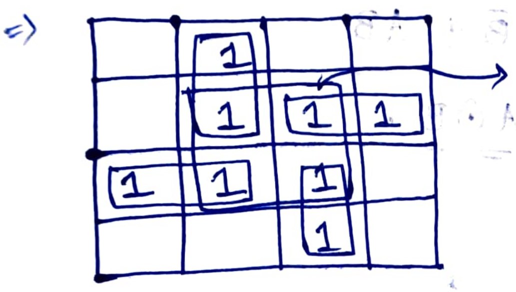


(Reduntant Pair)

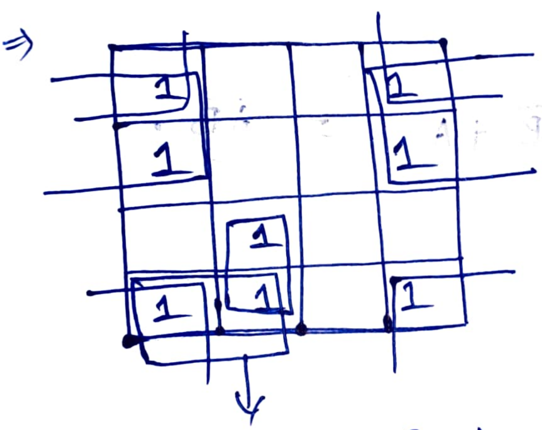


Redundant Pair

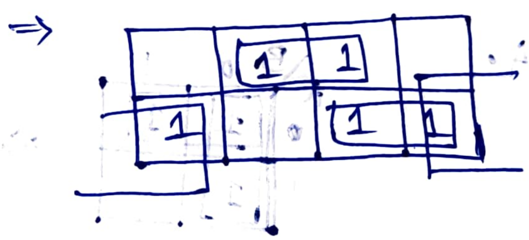
Redundant Pair



Redundant quad.



(Redundant Pair)



(v) effectively use don't care :- [X]

0	0	1	X	→ 1
0	0	1	1	

0	0	1	X	→ 0
0	0	1	0	

(vi) write down the boolean expression
[Target ⇒ Minimum Literal count]

(Q) $f(A,B) = \Sigma m(0,3)$. Write boolean expression.

A.

A \ B	0	1
0	1	
1		1

$\Rightarrow Y = \bar{A}\bar{B} + AB$
$= A \odot B$.

(Q) $f(A,B) = \Sigma m(0,1,2)$.

A.

A \ B	0	1
0	1	1
1	1	

$\Rightarrow Y = \bar{B} + \bar{A} = \overline{AB}$.

(Q) $f(A,B) = \pi M(1,2)$

A.

A \ B	0	1
0		0
1	0	

$\Rightarrow Y = (\bar{A}+B)\cdot(A+\bar{B})$.

(Q) $f(A,B) = \pi M(0,1,2)$

A.

A \ B	0	1
0	0	0
1	0	

$Y = A \cdot B = AB$.

(Q) $f(A,B) = \Sigma m(0,3) + \Sigma d(1)$

A.

A \ B	0	1
0	1	X
1		1

$Y = \bar{A} + B$

(Q) $f(A,B) = \pi M(0,2) \cdot \pi d(1)$

A.

A \ B	0	1
0	0	X → 1
1	0	

$Y = B$.

(Q) $f(A,B,C) = \Sigma m(2,3,5)$

A.

A \ BC	00	01	11	10
0			1	1
1		1		

$Y = B\bar{A} + A\bar{B}C$ //.

(Q) $f(A,B,C) = \Sigma m(0,2,3,4,6,7)$

A \ BC	00	01	11	10
0	1		1	1
1	1		1	1

$Y = B + \bar{C}$

(Q) $f(A,B,C) = \pi M(1,2,5,6)$

A.

A \ B+C	00	01	11	10
0		0		0
1		0		0

$Y = (\bar{B}+\bar{C})\cdot(\bar{B}+C)$

(Q) $f(A,B,C) = \pi M(0,2,3,6,7)$

A.

A \ B+C	00	01	11	10
0	0		0	0
1			0	0

$Y = (\bar{B})\cdot(A+C)$

(Q) $f(A,B,C) = \Sigma m(0,1,5,6) + \Sigma d(3,7)$

A.

A \ BC	00	01	11	10
0	1	1	X	
1		1	X	1

$Y = \bar{A}\bar{B} + C + AB$

(Q) $f(A,B,C) = \pi M(1,3,6,7)\cdot \pi d(2,4)$

A.

A \ B+C	00	01	11	10
0		0	0	X
1	X		0	0

$X \rightarrow 1$

$Y = \bar{B}\cdot(A+\bar{C})$

(Q) $f(A,B,C,D) = \Sigma m(0,1,3,4,5,6,7,13,15)$

A.

AB \ CD	00	01	11	10
00	1	1	1	
01	1	1	1	1
11		1	1	
10				

$Y = \bar{A}\cdot\bar{C} + \bar{A}\cdot D + B\cdot D + \bar{A}B$[illegible]

(Q) $f(A,B,C,D) = \Sigma m(1,5,6,7,11,12,13,15)$

A.

AB \ CD	00	01	11	10
00		1		
01		1	1	1
11	1	1	1	
10			1	

$Y = AB\bar{C} + \bar{A}\bar{C}D + ACD + \bar{A}BC$

(Q) $f(A,B,C,D) = \pi M(0,2,3,4,8,9,10,14)$

A.

A+B \ C+D	00	01	11	10
00	0		0	0
01	0			
11				0
10	0	0		0

$Y = (A+C+D)\cdot(\bar{A}+B+C)\cdot(A+B+\bar{C})\cdot(\bar{A}+\bar{C}+D)$

(Q) $f(A,B,C,D) = \pi M(1,3,5,8,9,12,14,15)$

A.

AB \ CD	00	01	11	10
00		0	0	
01		0		
11	0		0	0
10	0	0		

$Y = (A+B+\bar{D}) \cdot (A + C + \bar{D}) \cdot (\bar{A} + C + D) \cdot$
$(\bar{A} + B + C) \cdot (\bar{A} + \bar{B} + \bar{C})$

(Q) $f(A,B,C,D) = \Sigma m(4,5,7,12,14,15) + \Sigma d(3,8,10)$

A.

AB \ CD	00	01	11	10
00			X	
01	1	1	1	
11	1		1	1
10	X			X

$Y = B\bar{C}\bar{D} + \bar{A}BD + ABC$

(Q) $f(A,B,C,D) = \pi M(0,1,2,3,4,5) \cdot \pi d(10,11,12,13,14,15)$

A.

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	0		
11	X	X	X	X
10			X	X

$Y = (A+C)(A+B)$

(Q) You are given an n-variable logical expression. In K-map, you have done grouping of 2^k cells. What will be the literal count of that particular group?

[given :- $n > k$, $k \in$ Whole number]

(a) $2^n - 2^k$ (b) 0 (c) 1 (d) $n-k$

A. $n = 3$, $k = 1$

3 variables, $2^1 = 2$ grouping cells.

A \ BC				
			$\boxed{\ }$	

$Y = B\bar{C}$ L.C. $= 2$
$= 3 - 1$
$= n - k$

n	2^0	$n-0$
n	2^1	$n-1$
n	2^2	$n-2$
⋮	⋮	
n	2^k	$n-k$

(Q) K-map for $f(A,B,C,D)$ is given. Draw K-map for $f'(A,B,C,D)$

A.

AB \ CD	00	01	11	10
00		1	1	
01			1	
11		1	1	
10	1			1

$f(A,B,C,D) = \Sigma m(1,3,7,8,10,13,15)$

$f'(A,B,C,D) = \Sigma m(0,2,4,5,6,9,11,12,14)$

AB \ CD	00	01	11	10
00	1			1
01	1	1		1
11	1			1
10		1	1	

(Q) Map the given logical expressions in K-maps and find reduced logical expressions.

(i) $f(A,B,C) = \bar{A} + BC$

A \ BC	00	01	11	10
0	1	1	1	1
1			1	

$y = \bar{A} + BC$ (Same).

(ii) $f(A,B,C) = \bar{A} + AC$

A \ BC	00	01	11	10
0	1	1	1	1
1		1	1	

$y = \bar{A} + C$ //.

(iii) $f(A,B,C) = AC + BC$

A \ BC	00	01	11	10
0			1	
1		1	1	

$y = AC + BC$ (Same).

(iv) $F(A,B,C) = (A+C)(B+C)$ (POS).

A \ B+C	00	01	11	10
0			0	
1		0	0	

$y = (A+C)(B+C)$ (Same).

(v) $f(A,B,C,D) = ABC + CD + ABD$

AB \ CD	00	01	11	10
00			1	
01			1	
11		1	1	1
10			1	

$y = CD + ABC + ABD$ (Same).

(vi) $f(A,B,C,D) = \bar{A}(\bar{B}C + \bar{B}\bar{C} + BCD) + \bar{B}\bar{D}(A+C)$

$= \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} + \bar{A}BCD + A\bar{B}\bar{D} + \bar{B}C\bar{D}$

$d(A,B,C,D) = \bar{A}B\bar{C}D + \bar{A}BC\bar{D} + ACD$

AB \ CD	00	01	11	10
00	1	1	1	1
01		X	1	X → (1)
11			X	
10	1		X	1

$\therefore Y = \bar{A}\bar{B} + \bar{B}C + \bar{B}\bar{D}$ //

END OF

DAY - 10